A generalization for the Dirichlet beta function with connection to generalized Euler numbers using generalized hyperbolic functions

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Abstract

We introduce a family of generalized Dirichlet beta functions $\beta_n(s)$ defined for $n \in \mathbb{N}$ and $\mathrm{Re}(s) > 0$ via the normalized Mellin transform of the generalized hyperbolic secant. This construction recovers the classical Dirichlet beta function at $n = 2$, and produces a family of Barnes-type weighted zeta objects over $\mathbb{Z}[\zeta_n]$ for integer $n > 2$. we derive the identity $\beta_n(-mn) = \frac{(-1)^{mn}}{n} E_{mn}^{(n)}$ and characterize the trivial zeros of β_n.

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1 Introduction

The classical Dirichlet beta function is fundamentally linked to the hyperbolic secant via its regularized Mellin transform. In this paper, we generalize this framework by substituting the classical kernel with $\mathrm{sech_n}\,(x) = \frac{1}{\cosh_n\,(x)}$. This construction unifies canonical L-functions with weighted zeta objects over algebraic fields.

2 Prerequisite

Define Generalized $\cosh(x)$ as the "main cyclic function" of the n-th order as[4]:

$$\cosh_n\,(x) := \sum_{k=0}^{\infty} \frac{x^{nk}}{(nk)!} = \frac{1}{n} \sum_{m=0}^{n-1} e^{\omega^m x}, \quad \{x \in \mathbb{R}, n \in \mathbb{N}, \omega = e^{\frac{2\pi i}{n}}\}$$

Which is positive for all real valued x in $e^{\omega x}$
Define Generalized Euler numbers as [3]

$$E_{nk}^{(n)} = -\sum_{m=0}^{k-1} \binom{nk}{nm} E_{nk}^{(n)}$$

Define the Dirichlet beta function as [1]

$$\beta(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^s}$$

3 A closed form for $\mathrm{sech_n}(x)$ with multinational coefficients

We begin by defining the generalized hyperbolic secant natively for integer orders.
using the Euler form for $\cosh_n(x) = \frac{\sum_{k=0}^{n-1} e^{\omega^k x}}{n}$ we let

$$\mathrm{sech_n}(x) = \frac{n}{\sum_{k=0}^{n-1} e^{\omega^k x}}$$

Where $n \in \mathbb{N}$ and $\omega^n = 1$
Since $e^{\omega^0 x} = e^x$ we multiply by $\frac{e^{-x}}{e^{-x}}$ to get this infinite sum

$$\mathrm{sech_n}(x) = \frac{ne^{-x}}{1 + \sum_{k=1}^{n-1} e^{(\omega^k - 1)x}} = ne^{-x} \sum_{m=0}^{\infty} (-1)^m \left(\sum_{k=1}^{n-1} e^{(\omega^k-1)x} \right)^m$$

We let $A(x) = \left(\sum_{k=1}^{n-1} e^{(\omega^k-1)x} \right)$ to get this expression

$$\mathrm{sech_n}(x) = ne^{-x} \sum_{m=0}^{\infty} (-1)^m A(x)^m$$

However, we hit a boundary problem.
Around the origin, $A(0) = \sum_{k=1}^{n-1} 1 = n - 1$. This means that for $n > 2$, there exists a region where $|A(x)| > 1$.
Consequently, the geometric series expansion does not converge in the interval $[0, x_0]$, where x_0 is defined as the unique positive root of the equation $A(x_0) = 1$.
Beyond this point, for $x \geq x_0$, the condition $|A(x)| < 1$ holds.
To rigorously evaluate the integral across the entire domain, we must split it at x_0 and utilize the Taylor series definition of $\mathrm{sech_n}(x)$ for the interval $[0, x_0]$
Continuing outside this interval, the expansion for $A(x)^m$ equals

$$(\sum_{k=1}^{n-1} e^{(\omega^k-1)x})^m = \sum_{k_1+k_2+\ldots+k_{n-1}=m} \frac{m!}{k_1!k_2!\ldots k_{n-1}!} \prod_{j=1}^{n-1} (e^{(\omega^j-1)x})^{k_j}$$

We multiply the powers inside the finite product term $\prod_{j=1}^{n-1}(e^{(\omega^j-1)x})^{k_j} = \prod_{j=1}^{n-1}(e^{(\omega^j-1)xk_j}) = e^{x\sum_{j=1}^{n-1}k_j(\omega^j-1)}$ and including the e^{-x} factor inside we get $e^{-x}e^{x\sum_{j=1}^{n-1}k_j(\omega^j-1)} = e^{-x+x\sum_{j=1}^{n-1}k_j\omega^j-k_j} = e^{(-x+x\sum_{j=1}^{n-1}k_j\omega^j-x\sum_{j=1}^{n-1}k_j)}$
we know that $m = \sum_{j=1}^{n-1}k_j$ so the final shape of the exponent is $e^{-x(1+m-\sum_{j=1}^{n-1}k_j\omega^j)}$

$$\mathrm{sech_n}\,(x) = n\sum_{k_1+k_2+\ldots+k_{n-1}=m}(-1)^m\frac{m!}{k_1!k_2!\ldots k_{n-1}!}e^{-x(1+m-\sum_{j=1}^{n-1}k_j\omega^j)}$$

4 Generalized Dirichlet beta function

The Dirichlet beta function is defined as $\beta(s) = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^s}$, which is for the alternating Dirichlet character of period 4 has the integral representation

$$\beta(s) = \frac{1}{\Gamma(s)}\int_0^\infty \frac{x^{s-1}e^{-x}}{1+e^{-2x}}dx = \frac{1}{2\Gamma(s)}\int_0^\infty \frac{x^{s-1}}{\cosh(x)}dx \qquad \mathrm{Re}(s) > 0$$

We can see that the beta function is a Mellin transform of $\mathrm{sech}(x)$ normalized by a factor of $\frac{1}{2\Gamma(s)}$
Motivated by this structure, we define a generalized beta function associated with $\mathrm{sech_n}\,(x)$ by

$$\beta_n(s) := \frac{1}{n\Gamma(s)}\mathcal{M}\{\mathrm{sech_n}\,\}(s) \qquad \mathrm{Re}(s) > 0$$

for $n = 1, 2$ we get

$$\beta_1(s) = \frac{1}{\Gamma(s)}\mathcal{M}\{\frac{1}{e^x}\}(s) = \frac{1}{\Gamma(s)}\Gamma(s) = 1 \qquad\qquad \beta_2(s) = \beta(s)$$

and For $n = 2$ we have the classical Dirichlet $\beta(s)$ function, how ever for $n > 2$ the $\mathrm{sech_n}\,(x)$ doesn't converge between the interval $[0, x_0]$ as stated above
Instead, let's split the integral and use the series definition of $\mathrm{sech_n}\,(x)$
Let $I_1(s) = n\int_0^{x_0}\sum_{m=0}^{\infty}\frac{E_{mn}^{(n)}x^{s-1+nm}}{(nm)!}dx = n\sum_{m=0}^{\infty}\frac{E_{mn}^{(n)}}{(mn)!}\frac{x_0^{s+mn}}{s+mn}$ (which has poles at $s = 0, -1, -2$ etc) and $I_2(s) = n\int_{x_0}^{\infty}x^{s-1}\sum_{k_1+k_2+\ldots+k_{n-1}=m}(-1)^m\frac{m!}{k_1!k_2!\ldots k_{n-1}!}e^{-x(1+m-\sum_{j=1}^{n-1}k_j\omega^j)}dx$,
let $\lambda = 1+m-\sum_{j=1}^{n-1}k_j\omega^j$ then $I_2 = n\sum_{k_1+\ldots+k_{n-1}=m}(-1)^m\frac{m!}{k_1!\ldots k_{n-1}!}\frac{\Gamma(s,\lambda x_0)}{\lambda^s}$
where $\Gamma(a, b)$ is the upper incomplete gamma function
With this, one can possess another formula for calculations for $n \in \mathbb{N}$

$$\beta_n(s) = \frac{1}{n\Gamma(s)}n\left[\sum_{m=0}^{\infty}\frac{E_{mn}^{(n)}}{(mn)!}\frac{x_0^{s+mn}}{s+mn} + \sum_{k_1+\ldots+k_{n-1}=m}(-1)^m\frac{m!}{k_1!\ldots k_{n-1}!}\frac{\Gamma(s,\lambda x_0)}{\lambda^s}\right]$$

$$= \frac{1}{\Gamma(s)}\left[\sum_{m=0}^{\infty}\frac{E_{mn}^{(n)}}{(mn)!}\frac{x_0^{s+mn}}{s+mn} + \sum_{k_1+\ldots+k_{n-1}=m}(-1)^m\left(\frac{m!}{k_1!\ldots k_{n-1}!}\right)\Gamma(s,\lambda x_0)\lambda^{-s}\right]$$

where $\lambda = 1 + m - \sum_{j=1}^{n-1} k_j \omega^j$
we solve for $n = 2$ which in this case $\omega = -1$

$$\beta_2(s) = \frac{1}{\Gamma(s)} \left[\sum_{m=0}^{\infty} \frac{E_{mn}^{(n)}}{(mn)!} \frac{0}{s + mn} + \sum_{k_1 = m} (-1)^m \binom{m!}{k_1!} \Gamma(s, 0) \lambda^{-s} \right] = \sum_{n=0}^{\infty} \frac{(-1)^m}{(2n+1)^s}$$

we get back the original $\beta(s)$ series definition, and for the case $n = 1$ there is no sum which leads to $\beta_1(s) = 1$
These objects are out of the definition of the Dirichlet L series as they don't possess the form of $\sum \frac{a_n}{n^s}$, but they are closer to the Barnes zeta function type that takes the form $\zeta(s, w|a_1, ..., a_N) = \sum_{a_1,...,a_N \geq 0} \frac{1}{(w + n_1 a_! + ... + n_N a_N)^s}$
They can be defined as "Weighted Zeta objects With connection to $\mathbb{Z}[\omega]$ and have the Dirichlet beta function in the case of $\beta_2(s)$"

5 Convergence of $\beta_n(s)$

To evaluate its convergence, we split the integral $I_n(s) = \int_0^\infty x^{s-1} \operatorname{sech_n}(x)\, dx$ into two domains:

$$I_n(s) = \int_0^1 x^{s-1} \operatorname{sech_n}(x)\, dx + \int_1^\infty x^{s-1} \operatorname{sech_n}(x)\, dx$$

For the zero case ($x \to 0^+$) From the power series expansion
$\operatorname{sech_n}(x) = \frac{1}{1 + \frac{x^n}{n!} + \frac{x^{2n}}{(2n)!} + ...}$ $x \to 0^+$, all higher-order polynomial terms vanish, returning $\lim_{x \to 0^+} \operatorname{sech_n}(x) = 1$
Therefore, the local behavior of the integrand near zero is dominated by the power term $x^{s-1} \operatorname{sech_n}(x) \sim x^{s-1}$ as $x \to 0^+$. The improper integral $\int_0^1 x^{s-1}\, dx$ converges if and only if the real part of the exponent is greater than -1.
Thus, we establish the fundamental requirement is $\mathrm{Re}(s) > 0$
For the Asymptotic Behavior at Infinity ($x \to \infty$)Using roots-of-unity formula

$$\operatorname{sech_n}(x) = \frac{n}{\sum_{k=0}^{n-1} e^{\omega^k x}}$$

where $\omega = e^{\frac{2\pi i}{n}}$. Extracting the principal $k = 0$ term ($\omega^0 = 1$) from the denominator

$$\sum_{k=0}^{n-1} e^{\omega^k x} = e^x + \sum_{k=1}^{n-1} e^{x \cos(\frac{2\pi k}{n})} e^{ix \sin(\frac{2\pi k}{n})}$$

For all $1 \leq k \leq n-1$, the real component satisfies $\cos(\frac{2\pi k}{n}) < 1$. Consequently, as $x \to \infty$, the e^x term exponentially dominates the entire sum.
This gives us the asymptotic limit $\operatorname{sech_n}(x) \sim ne^{-x} \quad$ as $x \to \infty$
Plugging this back into our upper-bound integral:

$$\int_1^\infty x^{s-1} \operatorname{sech_n}(x)\, dx \sim n \int_1^\infty x^{s-1} e^{-x}\, dx$$

Because exponential decay overpowers any polynomial growth x^{s-1}, this integral converges absolutely for all $s \in \mathbb{C}$.
Combining both boundaries and the $\frac{1}{n\Gamma(s)}$ factor, $\beta_n(s)$converges absolutely and uniformly in the half-plane $\mathrm{Re}(s) > 0$, establishing a well-defined holomorphic function in this domain.

6 Connection to Euler numbers via Mellin Poles

To establish the relationship between the generalized Dirichlet beta function and the generalized Euler numbers, we must evaluate $\beta_n(s)$ at negative integers. Because the power series of $\mathrm{sech_n}\,(x)$ contains zero coefficients for all powers that are not a multiple of n, applying Ramanujan's Master Theorem directly requires an interpolating function that vanishes on a regular subset of integers. By Carlson's Theorem, such a function violates Hardy's exponential growth condition.
Instead, we apply a change of variables to eliminate the zero-padding before evaluating the Mellin transform.
Start with the convergent integral definition:

$$\beta_n(s) = \frac{1}{n\Gamma(s)} \int_0^\infty x^{s-1} \,\mathrm{sech_n}\,(x) dx$$

Let $u = x^n, x = u^{\frac{1}{n}}$ and $dx = \frac{1}{n} u^{\frac{1}{n}-1} du$.
Substituting this into the integral

$$\beta_n(s) = \frac{1}{n\Gamma(s)} \int_0^\infty \left(u^{\frac{1}{n}}\right)^{s-1} \mathrm{sech_n}\,(u^{\frac{1}{n}}) \left(\frac{1}{n} u^{\frac{1}{n}-1}\right) du$$

Combine the exponents to get

$$\beta_n(s) = \frac{1}{n^2\Gamma(s)} \int_0^\infty u^{\frac{s}{n}-1} \,\mathrm{sech_n}\,(u^{\frac{1}{n}}) du = \frac{1}{n^2\Gamma(s)} \mathcal{M}\{f_n\}(\frac{s}{n})$$

By definition, the Maclaurin series of this transformed kernel $f_n(u) = \sum_{m=0}^{\infty} \frac{E_{mn}^{(n)}}{(mn)!} u^m$ is strictly dense, containing no zero-gaps
According to the classical properties of the Mellin transform, if a function $f_n(u)$ is smooth on the positive real axis, decays exponentially as $u \to \infty$, and possesses a Taylor expansion $\sum c_m u^m$ near the origin, its Mellin transform $\mathcal{M}\{f_n\}(z)$ extends to a meromorphic function in the complex plane.[2]
Its only singularities are simple poles located at the negative integers $z = -m$, and the residue at each pole corresponds exactly to the Taylor coefficient:

$$\mathrm{Res}\,(\mathcal{M}\{f_n\}(z), z = -m) = c_m = \frac{E_{mn}^{(n)}}{(mn)!}$$

We can now evaluate $\beta_n(s)$ at negative integers. Let $s \to -kn$ for some non-negative integer k. In this limit, the Mellin argument becomes $z = \frac{s}{n} \to -k$.

The local Laurent expansion of the Mellin transform around this simple pole is

$$\mathcal{M}\{f_n\}(\frac{s}{n}) \approx \frac{c_k}{\frac{s}{n} - (-k)} = \frac{n \cdot c_k}{s + kn}$$

The reciprocal Gamma function in the denominator of $\beta_n(s)$ also possesses a simple pole at $s = -kn$. Its classical Laurent expansion is

$$\Gamma(s) \approx \frac{(-1)^{kn}}{(kn)!(s + kn)}$$

Evaluating the limit for $\beta_n(-kn)$ by taking the ratio of these two expansions, we find that the linear singularity poles $(s + kn)$ cancel out

$$\beta_n(-kn) = \lim_{s \to -kn} \frac{1}{n^2} \frac{\frac{n \cdot c_k}{s+kn}}{\frac{(-1)^{kn}}{(kn)!(s+kn)}} = \frac{1}{n} c_k (kn)!(-1)^{-kn}$$

Substituting the coefficient $c_k = \frac{E^{(n)}_{kn}}{(kn)!}$ back into the equation, the factorials cancel, which gives

$$\beta_n(-kn) = \frac{(-1)^{kn}}{n} E^{(n)}_{kn}$$

Furthermore, this framework naturally characterizes the trivial zeros of the function.
Let $s = -kn + r$, where $0 < r < n$. The Mellin argument becomes $z = -k + \frac{r}{n}$. Because this is not an integer, $\mathcal{M}\{f_n\}(z)$ is completely analytic and evaluates to a finite constant.
However, $\Gamma(-kn + r)$ remains a simple pole. A finite numerator divided by a singular Gamma function yields zero, thus proving

$$\beta_n(-kn + r) = 0$$

by setting $k = 0$ we get

$$\beta_n(0) = \frac{1}{n}$$

establishing the sequence of trivial zeros for the generalized Dirichlet beta function.

7 Future work

A generalization to any order $\alpha \in \mathbb{C}$ using the connection $\cosh_\alpha(x) = E_\alpha(x^\alpha)$ where $E_\alpha(x^\alpha)$ is the Mittag Leffler function

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